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Field Mapping of Formation Complexity to Solve Drilling Challenges

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Abstract

Drilling through 8.375-in and 5.875-in curve and lateral sections in gas fields involves hard and abrasive formations with varied thickness and characteristics. These sections require multiple bit trips, influenced by factors like formation abrasion index, hardness, interbedded layers, and varying sandstone and siltstone percentages. The project's objective was to conduct field mapping to understand these variations and develop a solution to reduce the number of trips needed to complete the sections.

The proposed solution involved analyzing the formation characteristics of each well in the field. A comprehensive study was conducted, integrating thorough analysis of offset well data, including well logs (sonic, gamma-ray, density-neutron, etc.), formation tops, mud logs, rock mechanics, drilling parameters, drill bit records, and dull bit conditions. Wells were plotted on a contour map to identify areas with high abrasion index, high interbedded index, and formation hardness. This detailed analysis aimed to enhance understanding of the geological variations across the field and improve decision-making for drilling operations.

The target wells can now be visualized on the field map, which helps identify the formation complexity in nearby offset wells. By analyzing factors such as unconfined compressive strength, effective porosity, abrasion characteristics, and impact potential, we can determine one or more optimal bit types and their specific applications. With this information, we can select the most suitable bit design, customizing features like cutter material and bit profile to match the formation conditions. Additionally, optimizing drilling parameters, such as weight on bit, rotary speed, and flow rate, further improves overall performance, leading to more efficient drilling operations and reducing bit trips to complete the section.

Given that the wells in this application have long lateral sections, understanding the formation complexity along these laterals is crucial. This knowledge enables the implementation of geo-steering, ensuring the well stays in the optimal zone for maximum performance. As a result, this approach leads to faster well delivery times and an increased number of wells drilled annually.

Introduction

This paper provides an in-depth exploration of the operational challenges encountered while drilling extended lateral sections through formations predominantly composed of hard and abrasive sandstone and siltstone. A common assumption in such drilling operations is that the geological formation along the lateral section is relatively homogeneous, exhibiting minimal variability in its formation strength and other characteristics. However, the findings presented in this study challenge that assumption, as a comprehensive formation analysis conducted for this specific application revealed significant heterogeneity in formation characteristics along the lateral section. These variations in geological and petrophysical properties introduced a unique set of complexities, impacting the efficiency and performance of the drilling process, and necessitating innovative approaches to mitigate associated risks and optimize operational outcomes.

Upon the completion of the formation analysis, an extensive and systematic mapping process was initiated to accurately chart the wells on a detailed contour map. This meticulous approach enabled the identification, visualization, and interpretation of spatial variations in formation characteristics across different locations within the region of interest. The mapping exercise served as a critical tool for delineating zones with distinctive formation characteristics, offering valuable insights into the heterogeneity of the subsurface formations. By integrating the findings from the contour mapping with the results of the formation analysis, the study facilitated a deeper understanding of the spatial distribution and variability of key formation attributes. This, in turn, allowed for the development of more informed strategies for well trajectory planning, enhanced decision-making in well placement, and the formulation of tailored drilling approaches, including selecting optimum bit type and cutter technologies to address the unique challenges posed by specific formation characteristics encountered along the lateral sections.

Challenges

The lateral sections drilled in the context of the subject application are typically performed within an 8.375-inch hole size, with the length of the lateral sections extending from approximately 4,000 feet to 5,000 feet. These sections are drilled through highly challenging hard and abrasive sandstone and siltstone formations, which impose significant demands on drilling equipment and operational strategies. The inherently abrasive and hard nature of these formations results in substantial wear and tear on the drill bits, often necessitating multiple bit trips to successfully complete the lateral section. One of the most frequently observed dull conditions on the drill bits is the severe wearing of cutters, which is indicative of the intense abrasion and cutting forces experienced during the drilling process.

Furthermore, it was observed that even within the same field and the same geological formation, the number of bits required to complete a lateral section exhibited considerable variability from one well to another. This inconsistency highlights the presence of localized heterogeneities in formation characteristics, variations in operational practices, or differences in downhole drilling dynamics. Such variability not only complicates the planning and execution of drilling operations but also underscores the importance of employing adaptive strategies and advanced tools to optimize performance and mitigate the challenges associated with these complex drilling environments.

To investigate the reasons behind the variability in the number of drill bits required to complete lateral sections from one well to another, a comprehensive study was undertaken involving the analysis of 11 wells. These wells were carefully selected to provide a representative sample of the field and were drilled within what was ostensibly the same geological formation. The objective of the study was to thoroughly examine the differing formation characteristics encountered along the lateral sections of these wells, with a particular focus on identifying the factors contributing to the variations in drilling performance and bit consumption. By analyzing these variations, the study aimed to uncover critical insights into formation heterogeneity, operational challenges, and their impact on bit durability and efficiency.

The figure 1 below illustrates the number of drill bits utilized in each of the 11 wells analyzed, along with the average footage drilled by each bit. This visual representation highlights the variance in bit performance observed across the wells, providing a clear depiction of the differences in bit durability and efficiency from one well to another. The variability showcased in the figure underscores the influence of localized formation characteristics, operational factors, and drilling dynamics on bit performance, offering valuable insights into the challenges faced during lateral drilling in similar geological environments.

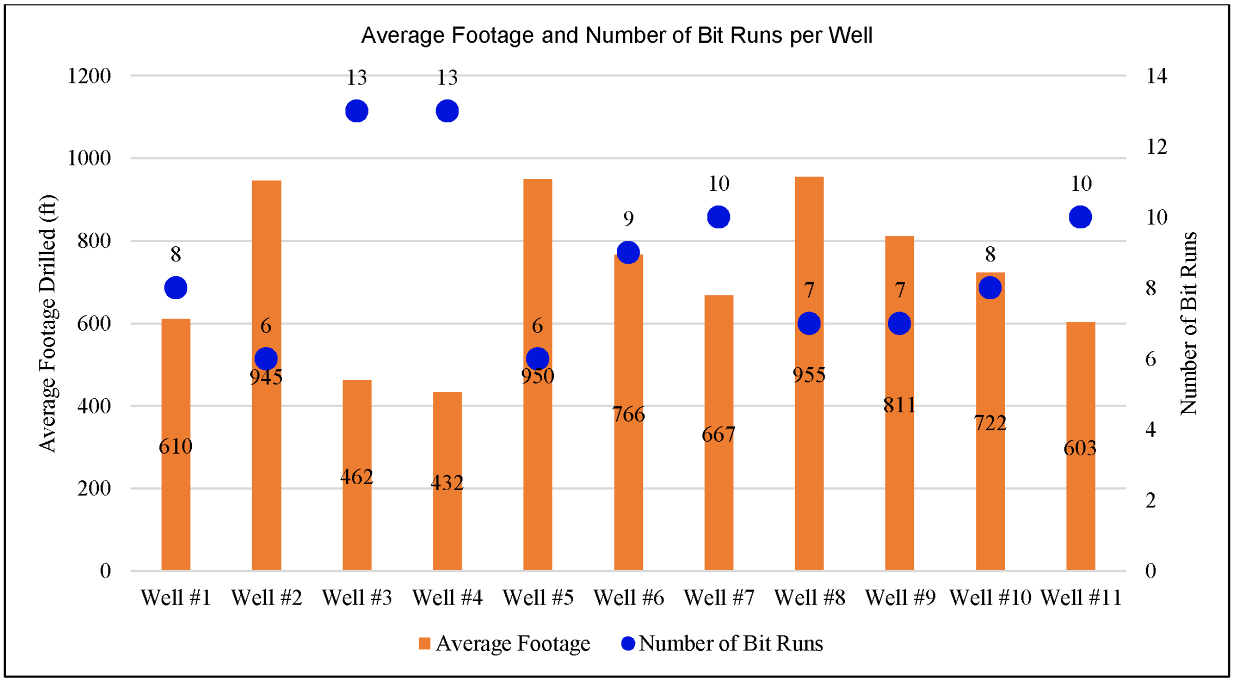


Figure 1—Average Footage and Number of Bit runs per Well

In order to better understand and systematically analyze the variation in bit performance observed across these wells, it was determined that a comprehensive formation analysis would be conducted. This analysis focused on key formation properties, including formation hardness, abrasiveness, and impact resistance, all of which are critical factors that can significantly influence bit performance during drilling operations. By evaluating these geological and mechanical characteristics in detail, the study aimed to identify the underlying causes of the observed variability in bit durability and efficiency, thereby providing a data-driven foundation for optimizing drilling strategies and improving performance outcomes in similar formations.

Formation Analysis Methodology

Formation analysis is a comprehensive and multidisciplinary study designed to gain a detailed understanding of the subsurface characteristics that influence drilling performance. This process involves the integration and thorough evaluation of offset well data, leveraging a wide range of inputs such as well logs—including sonic, gamma-ray, density-neutron, and other petrophysical measurements—along with formation tops, mud logs, and rock mechanics data. Additionally, drilling parameters, historical drill bit performance records, and dull bit conditions are carefully analyzed to correlate drilling dynamics with formation properties. By synthesizing these diverse datasets, the formation analysis provides critical insights into subsurface behavior, enabling more accurate predictions of formation challenges and facilitating the development of tailored drilling strategies to optimize performance.

Figure 2 below presents an example of the output logs generated from the formation analysis, providing a visual representation of the subsurface data and its interpretation. Using the input subsurface log data, the software generates detailed lithology profiles, capturing the variability in lithology and rock types across the analyzed interval. Furthermore, the analysis calculates and models the rock strength for the specified section, offering critical insights into the mechanical properties of the formation. The software also characterizes the formation's abrasiveness by evaluating the quartz content within the rock, a key factor influencing bit wear during drilling. In addition, the formation impact level is quantified based on the rate of variation in rock strength over the interval, providing a measure of the dynamic changes in mechanical properties. This integrated analysis enables a more precise understanding of the subsurface environment and its potential challenges, allowing for informed decision-making in bit selection and drilling operations.

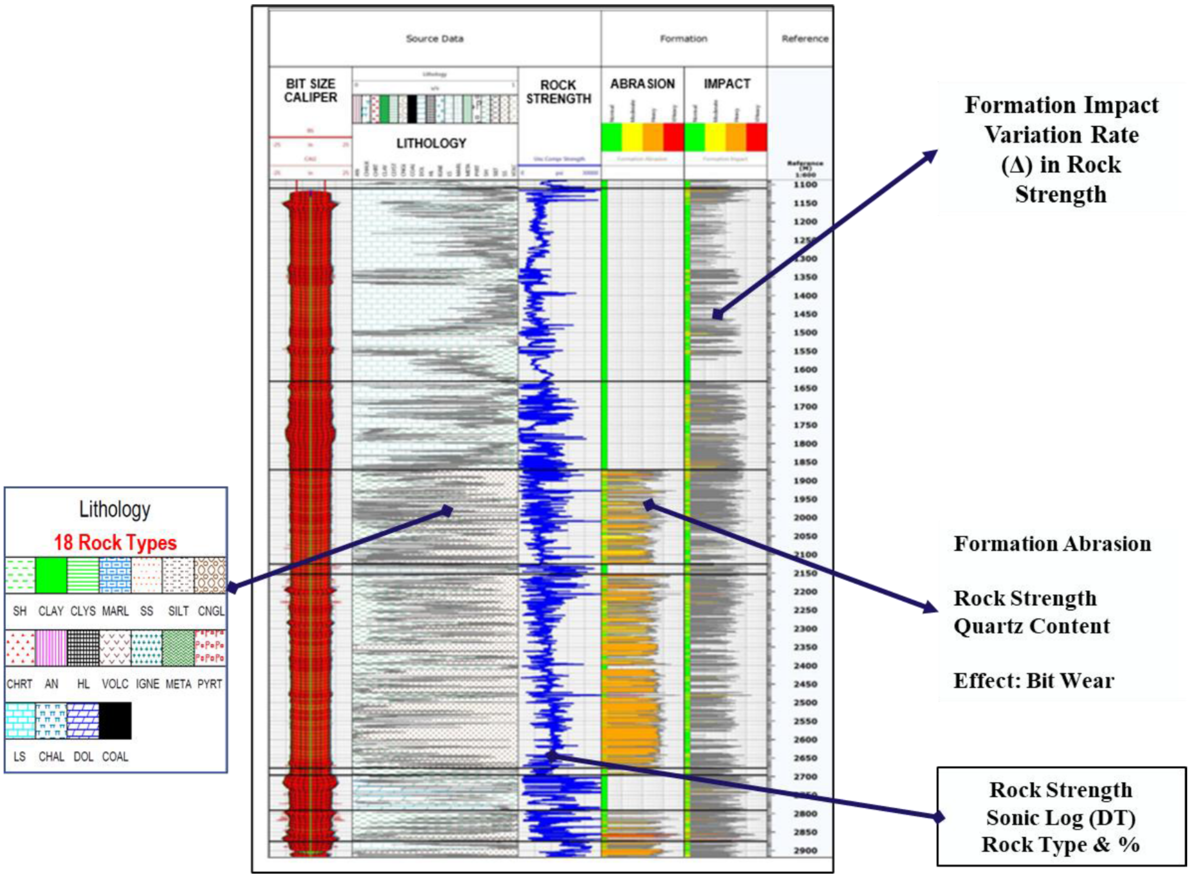


Figure 2—Formation Analysis and characterization of rock properties

Field Contour Mapping

Field contour mapping plays a critical role in identifying specific intervals within a formation of interest, providing a spatially resolved understanding of subsurface properties. By incorporating the geographic location of the proposed well, defined by its well coordinates, and comparing it to nearby offset wells, a comprehensive field map is developed. This map provides a visual representation of regional trends across the field, highlighting variations in formation characteristics and lithological distributions. Leveraging the insights from offset well data integrated into the field mapping, a synthetic formation analysis plot can be generated for the proposed well. This synthetic model enables the prediction of expected formation characteristics—such as rock strength, abrasiveness, and impact levels—along the planned well trajectory, facilitating proactive planning and optimization of drilling strategies tailored to the unique geological conditions anticipated in the target formation. An example of field contour mapping is shown below in figure 3.

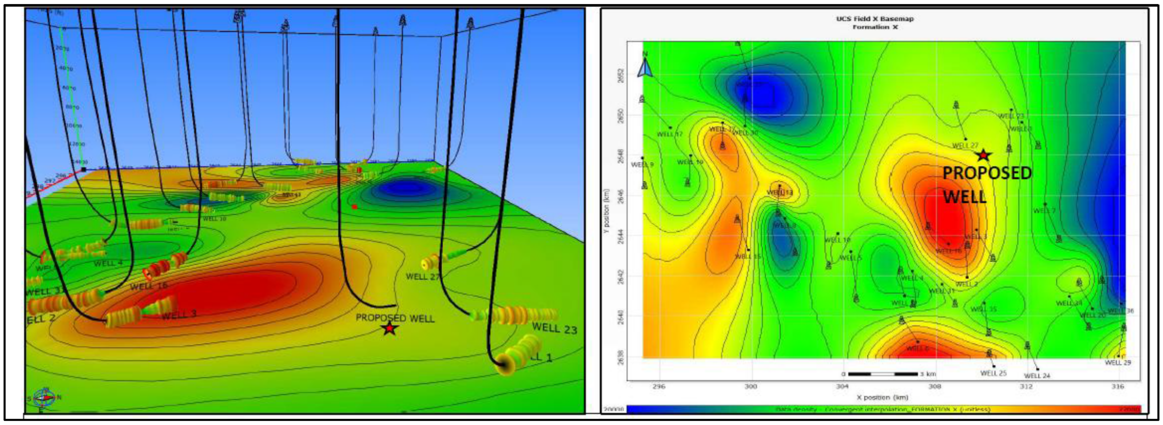


Figure 3—Featured example of contour mapping field study including offset lateral wells and the proposed well. Map color code based on rock strength.

Results of Formation Analysis Study

As part of the study conducted on 11 wells within the subject field, subsurface well logs and drill bit performance records were systematically collected, analyzed, and compared. A detailed formation analysis was performed for each well to evaluate the geological and mechanical properties encountered along the lateral sections. The resulting data and plots for each well were subsequently compared to identify trends and variations. Figure 4 below illustrates the evaluation of well data and drilling performance for the same formation type across the same field. The analysis revealed that, despite being within the same field, the formation characteristics of the same geological formations exhibited significant variability from one well to another. Notably, some wells that were located in close proximity to each other—often considered offset locations—still demonstrated substantial differences in formation properties, including variations in rock strength, abrasiveness, and impact levels. These findings underscore the inherent heterogeneity of the subsurface, even within a localized area, and highlight the importance of well-specific formation analysis for optimizing drilling performance and mitigating operational risks.

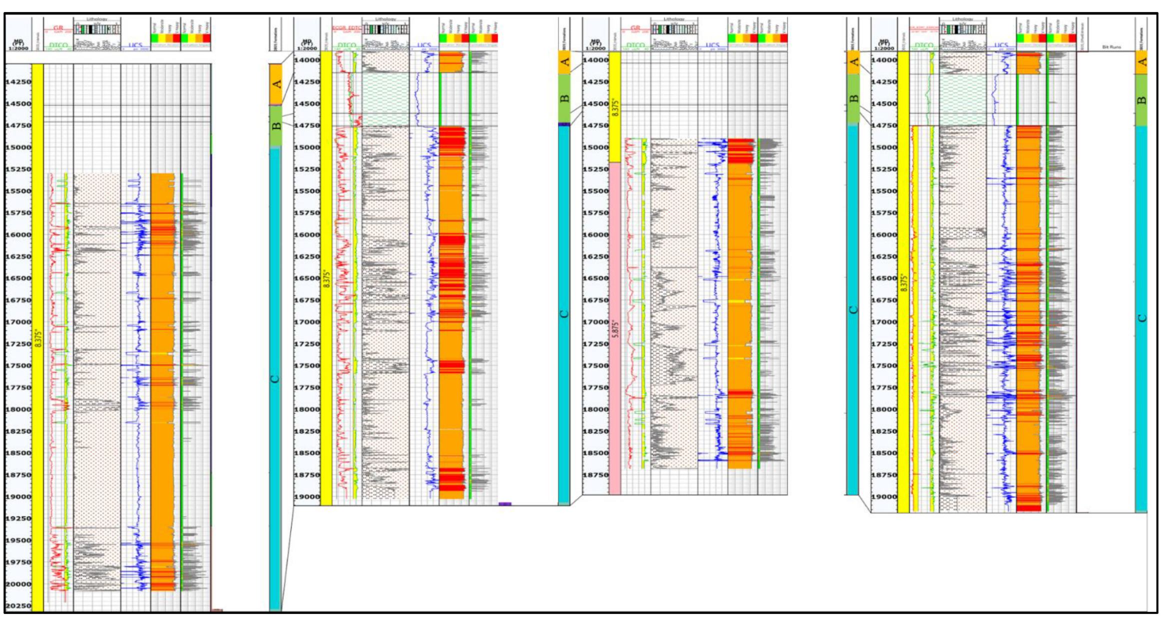
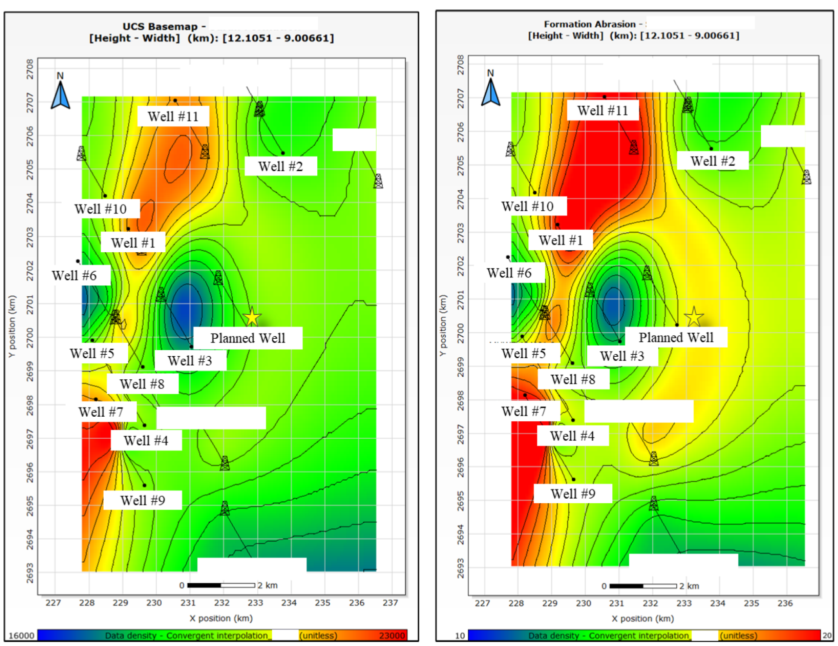


Figure 4—Formation characteristics comparison between well. Figure illustrates the variance in formation characteristics from one well to another

Results of Field Contour Mapping Study

As part of the study conducted on 11 wells within the subject field, a comprehensive field map was developed utilizing the geographic coordinates of the wells. This field map served as a critical tool for spatially visualizing and analyzing the distribution of formation characteristics across the field. The well locations were meticulously evaluated in conjunction with their corresponding formation properties, with particular emphasis on formation hardness and abrasiveness. This analysis enabled the identification of spatial trends and variances in geological properties, providing valuable insights into the heterogeneity of the subsurface environment. The field map not only facilitated a deeper understanding of the relationships between well locations and formation characteristics but also served as a foundation for optimizing drilling strategies and anticipating challenges in future well planning.

Figure 5 below presents the field contour mapping of the subject field, providing a visual representation of formation strength and abrasion levels across the area. The contour map employs a gradient color-coding scheme, with blue hues on the left end of the scale indicating regions characterized by lower formation strength and lower levels of abrasiveness. Conversely, red hues on the right end of the scale denote areas with higher formation strength and increased abrasiveness. This visual differentiation enables a clear understanding of the spatial variability in formation properties, allowing for the identification of distinct zones within the field. Such insights are instrumental in guiding the planning and execution of drilling operations, as they facilitate the anticipation of formation-specific challenges and the development of tailored operational strategies to optimize drilling performance.



Well Number	No of Runs	Formation Strength	Average Footage
Well #1	8 runs	21829 psi	610 ft
Well #2	6 runs	18802 psi	945 ft
Well #3	13 runs	18263 psi	462 ft
Well #4	13 runs	20326 psi	432 ft
Well #5	6 runs	19703 psi	950 ft
Well #6	9 runs	18556 psi	766 ft
Well #7	10 runs	20496 psi	667 ft
Well #8	7 runs	17940 psi	955 ft
Well #9	7 runs	19437 psi	811 ft
Well #10	8 runs	19394 psi	722 ft
Well #11	10 runs	21633 psi	603 ft

Figure 5—Field contour mapping corelating formation hardness and formation abrasiveness to the number of bits and average footage per bit for each well

The field mapping was subsequently correlated with the number of drill bits utilized in each well, enabling a comparative analysis of formation characteristics and bit performance. The results revealed a clear trend: well locations associated with higher formation strength and elevated abrasion values corresponded to an increased number of bit runs required to complete the lateral sections. This correlation underscores the direct impact of geological properties, such as formation hardness and abrasiveness, on drilling performance and bit durability. These findings provide critical insights into the relationship between subsurface heterogeneity and operational efficiency, emphasizing the importance of incorporating formation analysis and field mapping into well planning processes to optimize bit selection, mitigate tool wear, and improve overall drilling outcomes.

Bit Optimization based on Formation Analysis and Field Mapping

Based on the comprehensive data gathered and analyzed during the study, bit selection strategies can now be optimized by leveraging the geographic location of the planned well in conjunction with the formation characteristics derived from offset wells. By integrating insights from field contour mapping, formation analysis, and historical bit performance data, it is possible to predict the subsurface conditions likely to be encountered along the planned well trajectory. This predictive approach enables the selection of drill bits that are best suited to the anticipated formation hardness, abrasiveness, and impact levels, thereby enhancing bit performance, minimizing wear, and reducing the number of required bit trips. Consequently, this data-driven methodology not only improves drilling efficiency but also contributes to cost savings and operational reliability.

The most prominent dull condition observed on the drill bits was significant wear on the cutters, particularly in regions subjected to high work rates. To address this issue and improve bit durability, rolling cutter technology was strategically incorporated on the shoulder cutters, as this section of the bit endures the highest levels of mechanical stress and high work rates during drilling. The rolling cutter design effectively distributes wear across the cutting surface, thereby reducing localized stress concentrations and extending the operational life of the cutters in this critical area, ultimately enhancing overall bit performance. Furthermore, the cutting structures on the cone and nose areas of the bit were engineered with high thermal resistance cutters featuring thicker diamond tables. This advanced cutter design was specifically selected to mitigate thermal degradation—a common challenge in high-temperature drilling environments—and to withstand the abrasive nature of the formations encountered. By addressing both thermal and mechanical wear factors, these enhanced cutters are better equipped to maintain their integrity and cutting efficiency over extended intervals. The integration of these advanced design features has resulted in a drill bit optimized to endure the demanding conditions of drilling through hard and abrasive formations, leading to significantly improved durability, reduced bit trips, and enhanced operational efficiency.

Figure 6 below provides a visual representation of the prominent wear type observed on drill bits prior to the implementation of advanced cutter technologies (shown on the left). These wear patterns highlight the challenges associated with drilling through hard and abrasive formations. However, with the integration of the aforementioned cutter technologies—including rolling cutter designs on the shoulder and high thermal resistance cutters with thicker diamond tables on the cone and nose areas—the wear rates on the cutting structure have been significantly reduced.

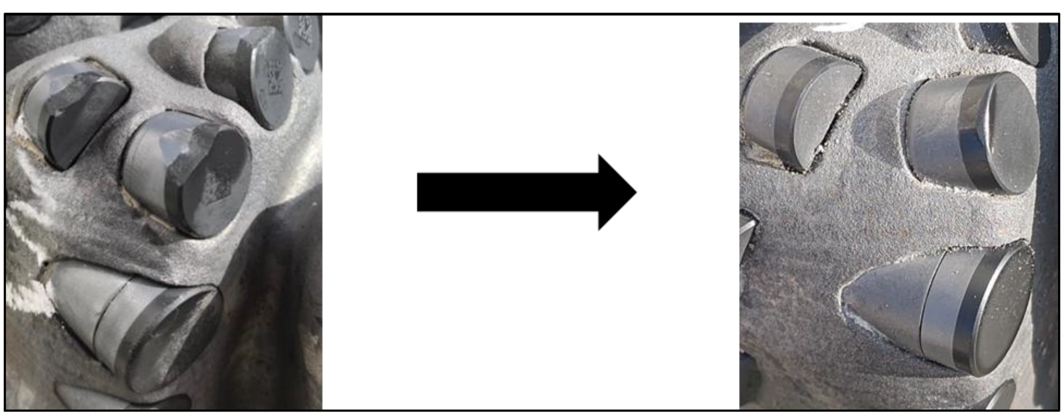


Figure 6—Dull condition comparison showing cutter wear before (on the left) and after (on the right) implementation of application specify cutters

Results

Following the implementation of formation characteristics analysis and comprehensive field mapping for the subject field, combined with meticulous planning, application-specific bit selection, and the integration of advanced cutter technologies, the results demonstrated significant improvements in operational efficiency. Specifically, there was a notable reduction in the number of bit trips required to complete the lateral sections, accompanied by an increase in the average footage drilled per bit. Furthermore, the application of these optimized strategies resulted in higher rates of penetration (ROP), effectively accelerating the drilling process. These advancements not only reduced the overall bit consumption, thereby lowering drilling costs, but also enabled the wells to be completed more quickly, translating into substantial time and cost savings for the operation. This outcome underscores the value of leveraging data-driven formation analysis and tailored bit design to enhance both performance and cost-efficiency in challenging drilling environments.

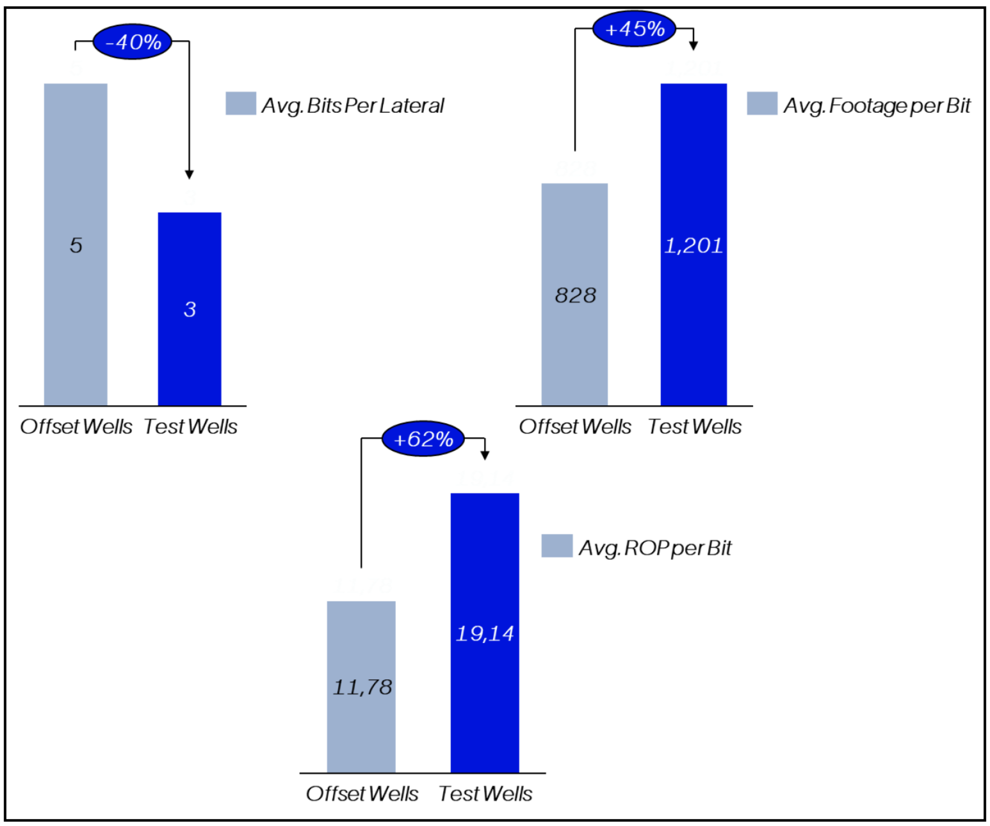


Figure 7—Test results illustrating reduction in bit trip and increase in operational efficiency

Conclusion

This paper highlights the significant value of leveraging data-driven formation analysis in conjunction with tailored bit design to optimize drilling performance and cost-efficiency in challenging subsurface environments. By integrating advanced geological analysis, field mapping, and application-specific bit technologies, the study demonstrates how these approaches can address operational challenges, reduce bit wear, minimize bit trips, and improve overall drilling efficiency. The findings underscore the importance of adopting a data-centric methodology to enhance the predictability and success of drilling operations, particularly in complex and abrasive formations.

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